

AERO40009 Thermodynamics and Heat Transfer
Tutorial Sheet 4

I: Revision problem (Chapters 3 & 4).

- Q1. (a)** A cylinder of 2 m³ capacity contains air at 300 K and 2 bar. The surrounding atmosphere is at a pressure of 1 bar and a temperature of 300 K.
- (i) What is the mass of the air?
 - (ii) What does the exergy of the air measure?
 - (iii) Calculate the exergy of the air.
- (b)** The air then undergoes a reversible, isothermal expansion to 1 bar pressure.
- (i) Calculate the work done.
 - (ii) Why is there a difference between the work done and the exergy?

(a) (i) $PV = MRT$

$$\therefore M = \frac{PV}{RT} = \frac{4000}{861} \text{ kg} = 4.646 \text{ kg}$$

(ii) Maximum amount of useful work that air in cylinder could perform in reaching equilibrium with its surroundings

(iii) $V_1 = V_a \frac{P_a}{P_i} = 4 \text{ m}^3$

$$X = M\phi = M(\Delta u + P\Delta v - T\Delta s)$$

$$= M \left[C_p \Delta T + P_a \frac{\Delta V}{M} - T_a \left[C_v \ln\left(\frac{T_a}{T_i}\right) + R \ln\left(\frac{V_a}{V_i}\right) \right] \right]$$

$$= P_a \Delta V - MRT_a \ln\left(\frac{V_a}{V_i}\right)$$

$$= 1 \times 10^5 (2 - 4) - \frac{4000}{861} \cdot 287 \cdot 300 \cdot \ln\left(\frac{2}{4}\right)$$

$$= 77258.87222 \text{ J}$$

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$$\begin{aligned} \text{(b) (i)} \quad W &= \int P dV = \int \frac{MRT}{V} dV \\ &= MRT \int \frac{1}{V} dV \\ &= MRT \ln \left(\frac{V_1}{V_a} \right) \\ &= \frac{4000}{861} \cdot 287 \cdot 300 \cdot \ln \left(\frac{4}{2} \right) \\ &= 277258.8722 \text{ J} \end{aligned}$$

$$\begin{aligned} \text{ii} \quad W - X &= \int P_a dV \\ &= P_a \Delta V \\ &= \text{Work done in displacing the surroundings} \end{aligned}$$

II: Gas dynamics.

- Q2. A perfect gas is expanded isentropically from a reservoir where the pressure is P_o , density ρ_o and the absolute temperature T_o .

Obtain expressions for the pressure, density and temperature ratios P/P_o , ρ/ρ_o and T/T_o at a throat where $M = 1$.

Calculate their ratios for $\gamma = 1.4$

$$\frac{1}{2}u^2 + C_p T = \text{constant} = h_o$$

$$\begin{aligned}\therefore \frac{1}{2}(Ma)^2 + C_p T &= \frac{1}{2}M^2 \gamma R T + C_p T \\ &= \frac{1}{2}M^2 \gamma (C_p - C_v) T + C_p T = \text{constant} = C_p T_o\end{aligned}$$

$$\therefore \frac{1}{2}M^2 \gamma \left(1 - \frac{1}{\gamma}\right) T + T = T \left(1 + \frac{\gamma-1}{2} M^2\right) = T_o$$

$$\therefore T = \frac{P}{\rho R} \quad \therefore \frac{P}{\rho} \left(1 + \frac{\gamma-1}{2} M^2\right) = \frac{P_o}{\rho_o}$$

$$\therefore P \gamma^{\frac{1}{\gamma}} = \frac{P}{\rho^{\frac{\gamma}{\gamma-1}}} = \text{constant} \quad \therefore P = \text{const} \cdot \rho^{\frac{\gamma}{\gamma-1}} \quad \therefore \rho = \frac{P^{\frac{\gamma-1}{\gamma}}}{\text{constant}}$$

$$\therefore \rho^{\gamma-1} \left(1 + \frac{\gamma-1}{2} M^2\right) = \rho_o^{\gamma-1} \quad \therefore \rho \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{1}{\gamma-1}} = \rho_o$$

$$\therefore P^{1-\frac{1}{\gamma}} \left(1 + \frac{\gamma-1}{2} M^2\right) = P_o^{1-\frac{1}{\gamma}} \quad \therefore P \left(1 + \frac{\gamma-1}{2} M^2\right)^{\frac{\gamma}{\gamma-1}} = P_o$$

When $M = 1$, $\gamma = 1.4$,

$$\frac{T_o}{T} = 1.2 \quad \therefore \frac{T}{T_o} = 0.83, \quad \frac{P}{\rho_o} = 0.63, \quad \frac{P}{P_o} = 0.53$$

- Q3.** Show that the maximum mass flow rate (choked flow) through a convergent - divergent nozzle with a throat area A_T can be expressed as:

$$\dot{m} = A_T \left(\gamma P_0 \rho_0 \left[\frac{2}{\gamma+1} \right]^{\frac{\gamma+1}{\gamma-1}} \right)^{\frac{1}{2}}$$

or in terms of T_0 and P_0 as:

$$\dot{m}^* = A_T \frac{P_0}{\sqrt{T_0}} C$$

where

$$C = \sqrt{\gamma} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

and P_0 , T_0 and ρ_0 are the reservoir pressure, temperature and density respectively.

$$\dot{m} = \rho^* u^* A_T$$

$$\text{At } M = 1$$

$$u^* = Ma = a = \sqrt{\gamma R T^*} = \sqrt{\gamma \frac{P^*}{\rho^*}}$$

$$\therefore \dot{m} = \rho^* \sqrt{\gamma \frac{P^*}{\rho^*}} A_T = \sqrt{\gamma} \sqrt{P^* \rho^*} A_T$$

$$\therefore \frac{P_0}{P} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}} \quad \therefore P^* = P_0 \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}}$$

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{1}{\gamma-1}} \quad \therefore \rho^* = \rho_0 \left(\frac{2}{\gamma+1} \right)^{\frac{1}{\gamma-1}}$$

$$\therefore \dot{m} = \sqrt{\gamma} \sqrt{P_0 \rho_0} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} A_T$$

- Q4.** The Imperial College hypersonic gun tunnel may be regarded as a blow-down wind tunnel in which air expands through a convergent-divergent nozzle of circular cross-section from a reservoir where the pressure is 586 bar and the temperature is 1100 K.

If the temperature in the test jet, downstream of the nozzle is not to fall below 60K, what is the maximum Mach number to which the flow can be expanded and what is then the pressure in the jet?

If the diameter is 24cm, what must be the throat diameter for this maximum Mach number?

$$T_0 = T \left(1 + \frac{\gamma - 1}{2} M^2 \right)$$

$$\therefore 1100 = 60 \left(1 + \frac{1.4 - 1}{2} M^2 \right)$$

$$\therefore M_{max} = 9.31$$

$$\therefore \frac{P_0}{P_{min}} = \left(1 + \frac{\gamma - 1}{2} M_{max}^2 \right)^{\frac{\gamma}{\gamma - 1}}$$

$$\therefore P_{min} = 2221 \text{ Pa}$$

$$\begin{aligned} \therefore \dot{m} &= \rho u A = \frac{P_{min}}{RT} \sqrt{\gamma RT} M \pi \left(\frac{d}{2} \right)^2 \\ &= 8.43 \text{ kg/s} \end{aligned}$$

At throat,

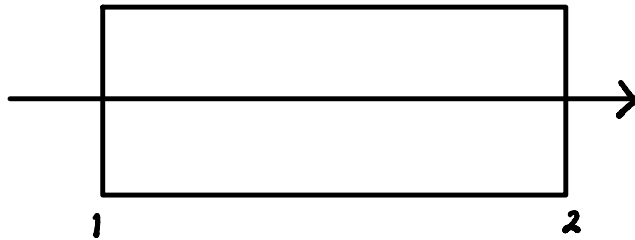
$$\dot{m} = A_T \frac{P_0}{\sqrt{T_0}} \sqrt{\frac{\gamma}{R}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma + 1}{2(\gamma - 1)}} = 8.43 \text{ kg/s}$$

$$\therefore D_T = 1.226 \text{ cm}$$

- Q5. A perfect gas flows along a frictionless pipe of constant area.
By considering the momentum and continuity equations, show that

$$\frac{T^{\frac{1}{2}}}{M} (1 + \gamma M^2) = \text{constant}$$

D J Doorly (rev. 2023)



$$\text{Continuity : } \rho_1 u_1 A_1 = \rho_2 u_2 A_2 \quad \therefore \rho u A = \text{constant}$$

$$\text{Momentum : } P_1 + \rho_1 u_1^2 = P_2 + \rho_2 u_2^2 \quad \therefore P + \rho u^2 = \text{constant}$$

$$\therefore A \text{ do not change} \quad \therefore \rho u = \text{constant}$$

$$\therefore \frac{P}{\rho u} + u = \text{constant} \quad \therefore P = \rho R T, \quad u = M a = \sqrt{\gamma R T} M$$

$$\therefore \frac{R T}{\sqrt{\gamma R T} M} + \sqrt{\gamma R T} M = \text{constant} \quad \therefore R = \text{constant}$$

$$\therefore \frac{\sqrt{T}}{\sqrt{\gamma} M} + \sqrt{\gamma T} M = \text{constant} \quad \therefore \gamma = \text{constant}$$

$$\therefore \frac{\sqrt{T}}{M} (1 + \gamma M^2) = \text{constant}$$